Geothermal Energy

Exam code: 0625 & 0972

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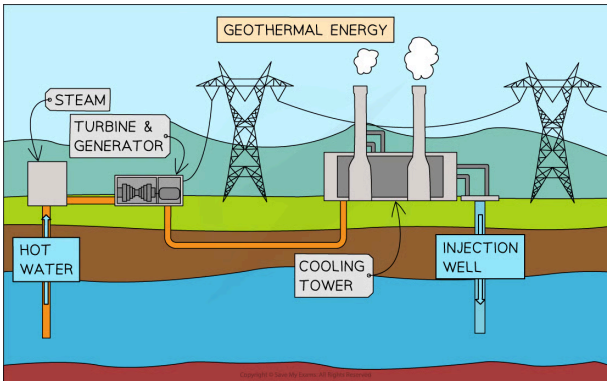
Geothermal energy

- **Tectonic activity** causes areas where the rocks below ground become **heated**
- Energy in the thermal store of the rocks can be obtained by:
  - Drilling **shafts** into the Earth's surface to expose the heated rocks
  - Injecting **water** into the shaft which is heated by the rocks
  - The water is returned via another shaft as **steam** or hot water
  - The steam can be used to turn a **turbine** and generate electricity
  - The hot water can be used to heat homes and provide a **hot water** supply to households

Geothermal power station



Cold water is heated by the rocks and returned as hot water or steam, which can be used to generate electricity

Advantages of geothermal energy

- Geothermal energy is **renewable**
- Geothermal power stations are **reliable** because the rocks remain hot for thousands of years so they can consistently produce electricity

Disadvantages of geothermal energy

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

and can be forced out when water is injected into the rocks (fracking)
    - Methane is a **greenhouse gas** which contributes to global warming (even without combustion)

- Geothermal power stations are **expensive** to build
- Geothermal power stations are **small** so they are not suitable for large-scale electricity production
- Geothermal power stations are **not very efficient** (around 12%)

Examiner Tips and Tricks

In your IGCSE exam, you may be required to discuss the advantages and disadvantages of various energy resources. You may also need to apply your understanding of energy resources to unfamiliar scenarios, for example, by suggesting which type of energy resource would be best for a particular community. Make sure you can justify your choices.

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